

Conglomerate Mergers: Effects on Growth and Competition

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Motivation



- A conglomerate is a combination of business units operating in entirely different industries
- Usually they span multiple countries (multinational)
- There are more than 1,000 conglomerates worldwide
- Highly dynamic process of merging, acquiring and splitting
- Many of the brands you use daily probably belong to a conglomerate

Key Questions

- 1 Why would firms in *unrelated* markets merge?
- 2 How could antitrust authorities evaluate conglomerative mergers?
- 3 How do management costs affect conglomerate structure?

Seeking Your Input

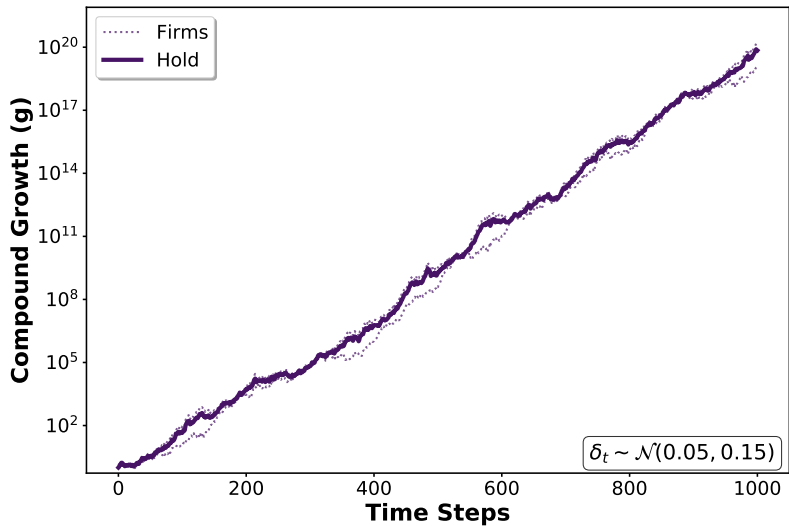
Publication Strategy Discussion

I would greatly appreciate your thoughts on:

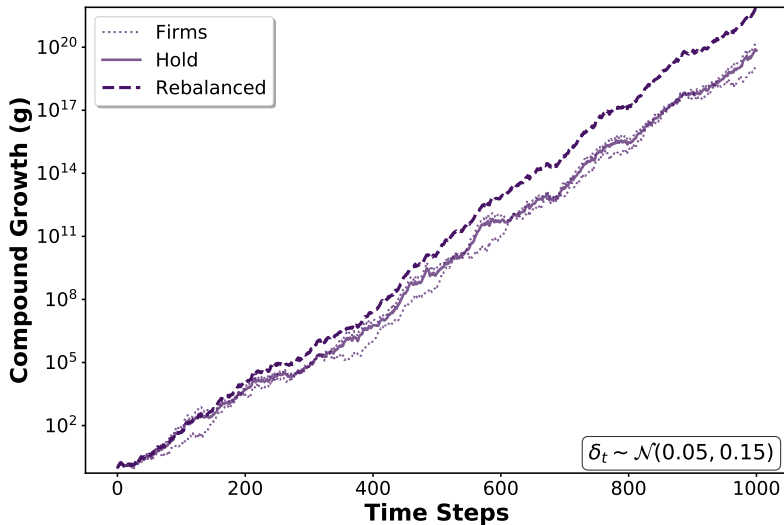
- Where this work fits within the existing literature
- Which journals or outlets might be most suitable
- What modifications would strengthen the contribution for publication

Your expertise and suggestions would be invaluable!

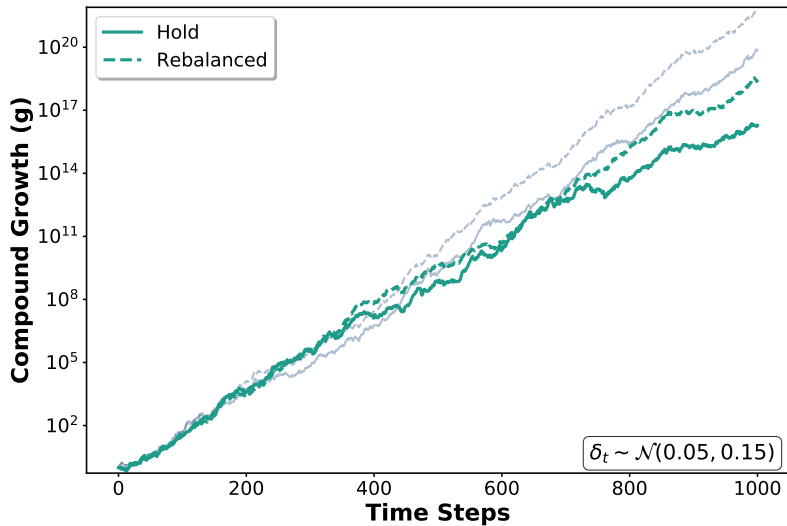
The Power of Resource Pooling



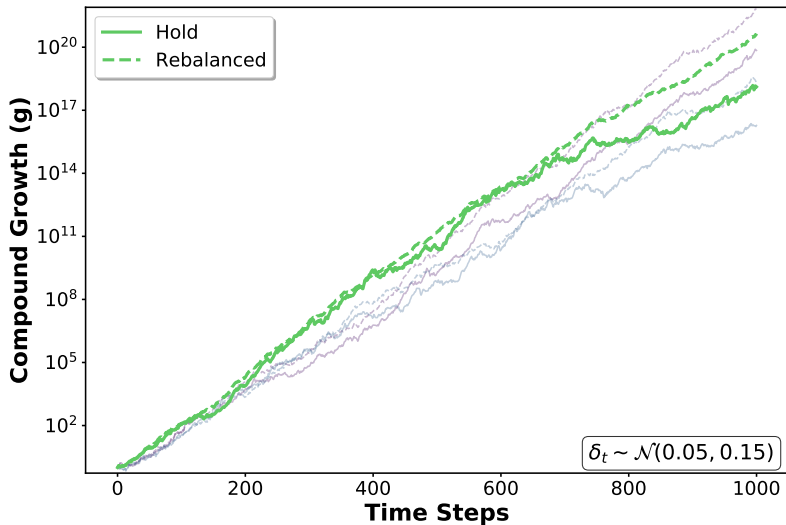
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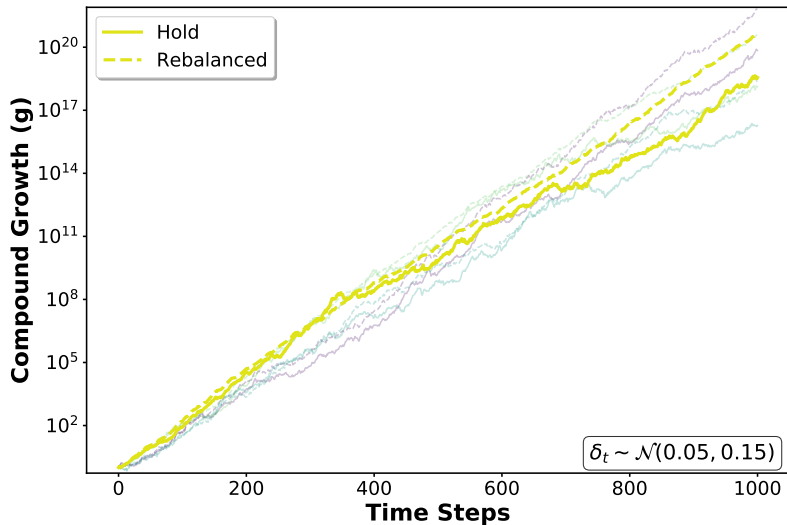
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Multiplicative Growth

- $g = \prod_{t=1}^T \delta_t$ instead of $g = \sum_{t=1}^T \delta_t$
- “When tomorrow depends on today” $\rightarrow s_{t+1} = s_t \delta_t$
- All economic dynamics that are denoted in % are usually linked to multiplicative dynamics
- Prevalence of heavy tailed distributions in many economic domains well explained by multiplicativity
- Examples of multiplicative processes
 - Capital accumulation
 - Trust, reputation
 - Legal cases
- Investment in capital goods don't just add capacity, it transforms the entire production structure

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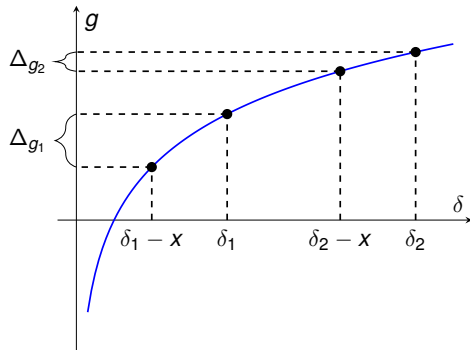
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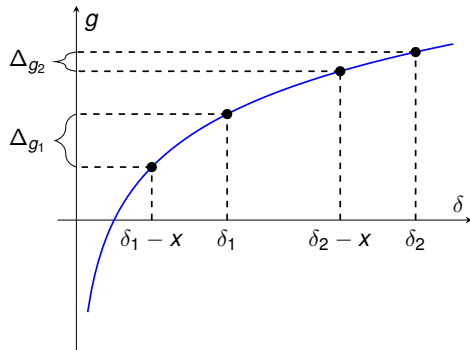
Growth and Risk: Two Sides of the Same Coin

- Losses hurt exponentially more, while gains yield exponentially less ($\Delta_{g_1} > \Delta_{g_2}$)
- Risk $\neq$ Volatility, Risk = Loss (large upswings aren't as important)
- Irreversibility of state transitions ($\delta_t = 0$) makes things even worse (think Russian Roulette)
- Shifting the return distribution upwards, avoiding very small growth increments



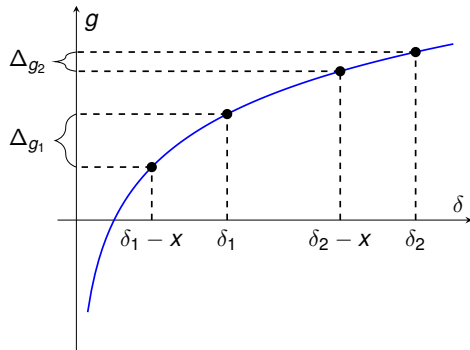
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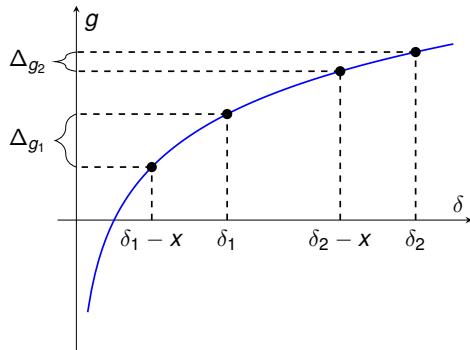
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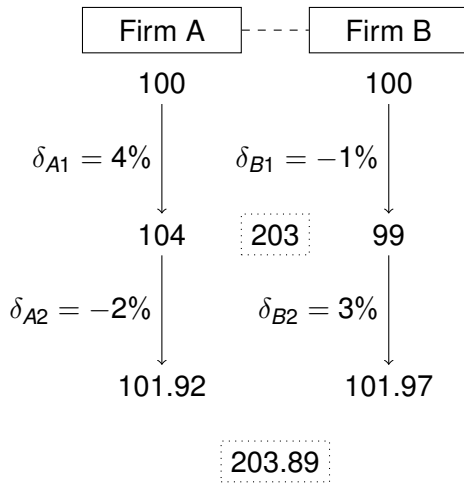
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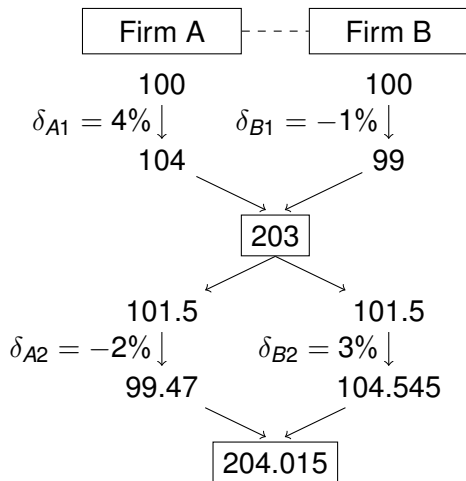
Sharing is Sparing

- Sharing (pooling) of resources reduces the impact of a loss $\rightarrow$ insurance
- Peters and Adamou (2022) show that with sharing under Gaussian growth increments we can increase $g = \mu - \frac{\sigma^2}{2}$ to $g = \mu - \frac{\sigma^2}{2N}$, with N the number of sharing partners
- Giving up resources to reduce a loss can be worth more than the resources themselves (positive sum)

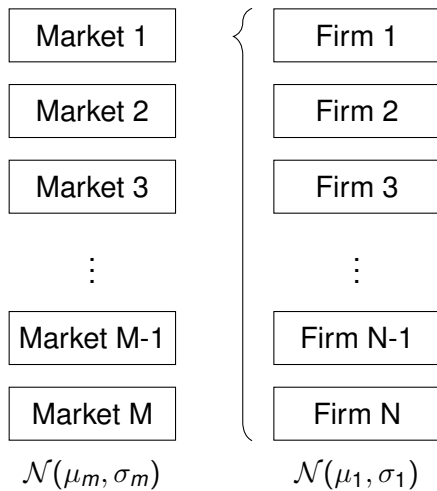


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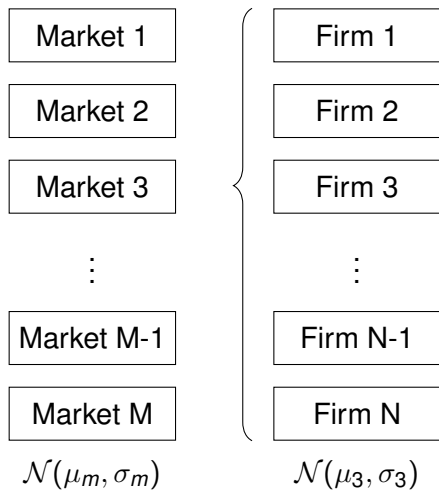
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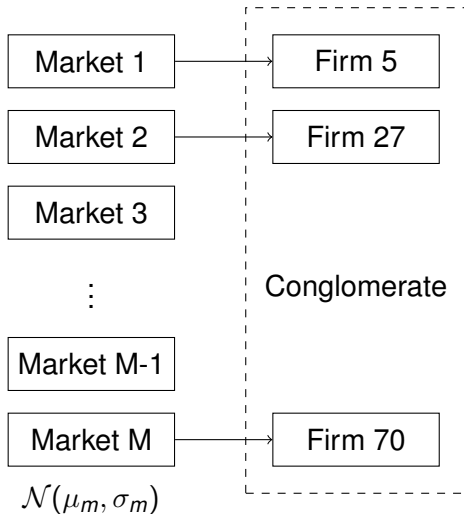
The Anatomy of an Agent Based Model



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Growing in the Markets

$$\text{Firm}_{im} \quad \delta_{it} \sim \mathcal{N}(\mu_m, \sigma_m)$$

$$\Delta_{it} = s_{it-1} \delta_{it} - s_{it-1}$$

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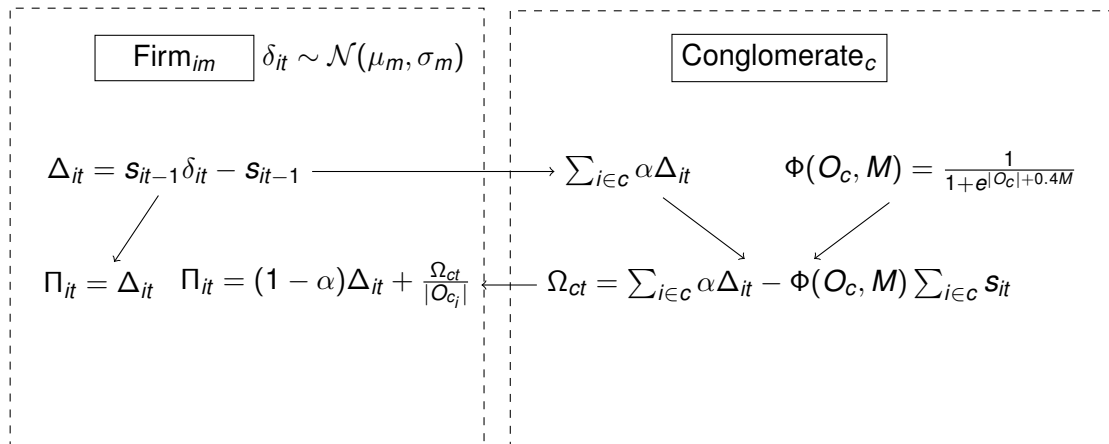
Conglomerate_c

$$\sum_{i \in c} \alpha \Delta_{it}$$

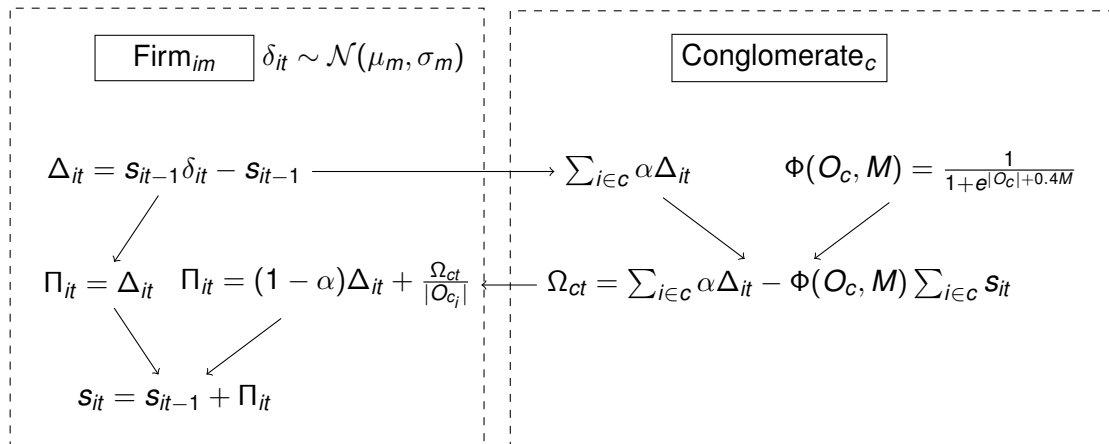
$$\Phi(O_c, M) = \frac{1}{1 + e^{|O_c| + 0.4M}}$$

$$\Omega_{ct} = \sum_{i \in c} \alpha \Delta_{it} - \Phi(O_c, M) \sum_{i \in c} s_{it}$$

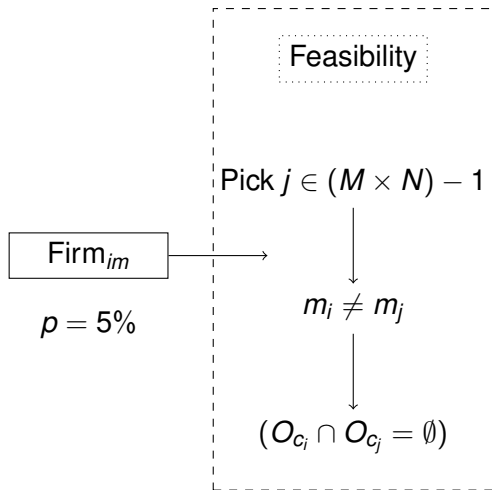
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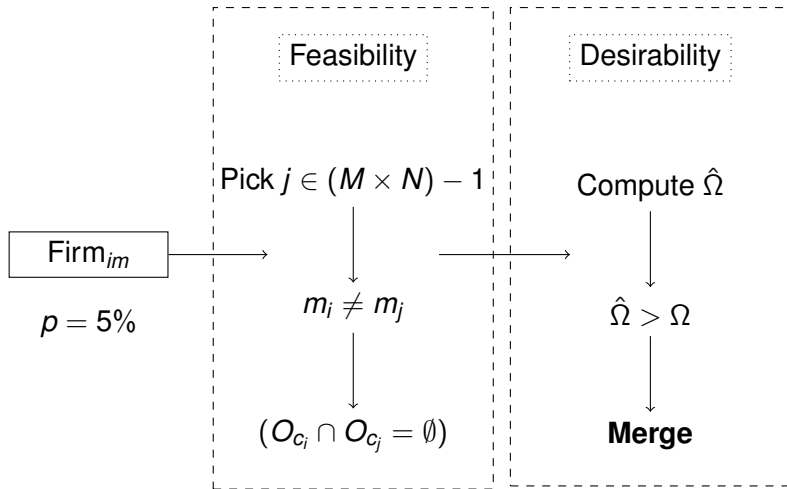
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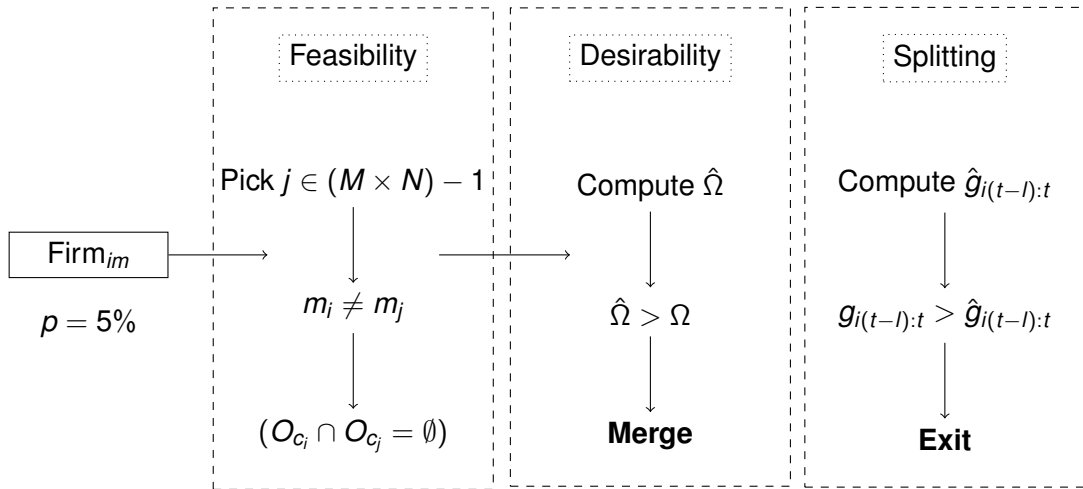
Merging and Splitting



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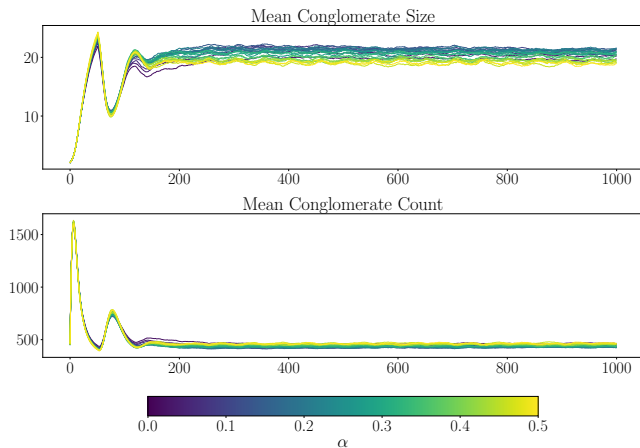


Parametrization and Experimental Setup

Parameter	Value
M	100
N	100
T	1000
I	50
p	0.05
α	[0, 0.5]

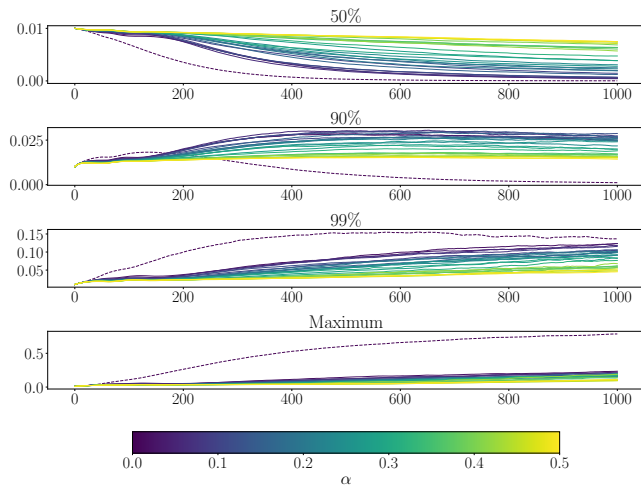
- 100 Markets means maximum conglomerate size is 100
- Increase the sharing rate α by 0.02
- Run each parametrization 10 times and average along their trajectories
- Management cost function set such that operating in 40 % of the markets cost 50 % of the conglomerate each step

Merging Behavior



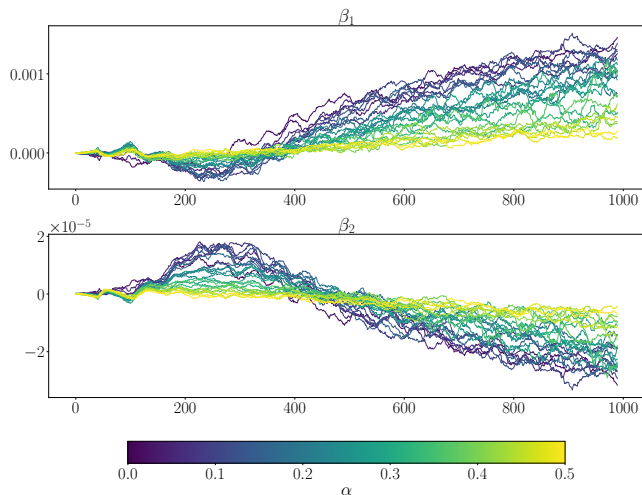
- Stable conglomerate size
- Stable number of conglomerates
- Stabilization after the initial / review periods
- Increasing α decreases conglomerate size
- Increasing α marginally decreases conglomerate count

The Impact of the Pooling Rate on Market Shares



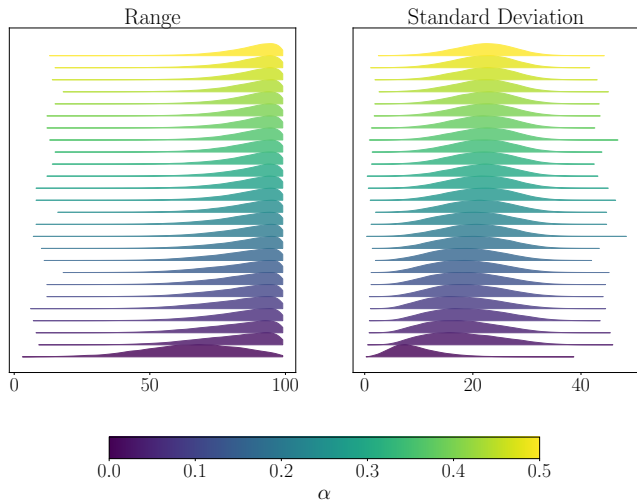
- Perfect equality is 0.01 with 100 firms per market
- Inequality increases with time
- Non-pooling returns the expected log-normal distribution
- Increasing α decreases market share inequality and slows increase over time

Conglomerate Size and Market Share



- Impact of conglomerate size on market share of firms within the conglomerate
$$\bar{\Omega}_{ct} = \beta_{0t} + \beta_{1t}|O_{ct}| + \beta_{2t}|O_{ct}|^2$$
- $\beta_1 > 0$: Variance reduction
- $\beta_2 < 0$: Management cost
- Increasing α decreases the effects as market shares tend to equalize

The Impact of the Pooling Rate on Mobility



- Range as the difference between ranks over time
- Standard deviation of ranks over time
- Any $\alpha > 0$ improves mobility drastically
- Increasing α shifts distribution to higher mobility
- Increasing α decreases the spread of distribution

Key Findings

- 1 Growth rate optimization is a novel view in standard models of industrial organisation
- 2 Growth optimization through resource sharing is a possible reason why conglomerates form
- 3 Pooling of resources within the conglomerate allows for improved performance for all companies with it
- 4 Increasing firm cooperation through conglomerate formation improves static and dynamic measures of competitiveness
- 5 Management costs play a crucial role in determining the size of a conglomerate

Future Developments

- 1 Other resources outside of capital could be modelled
- 2 Include jump processes or other return distributions
- 3 Robust management cost function for conglomerate size
- 4 More guided merger targeting process
- 5 Investigate how conglomerates change barriers to entry

References I

Peters, O. and Adamou, A. (2022). The ergodicity solution of the cooperation puzzle. *Philosophical Transactions of the Royal Society A*, 380(2227):20200425.

Reach Out!

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